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1 import tkinter as tk
2 from tkinter import messagebox
3 import database
4 import logic
5
6 class KioskGUI:
7     def __init__(self, root):
8         self.root = root
9         self.root.title("Salone Kiosk Ledger - CSV
Edition")
10        self.root.geometry("600x650")
11        self.root.configure(bg="#f4f6f9")
12
13        # Initialize the CSV file on startup
14        database.init_db()
15
16        self.build_ui()
17
18    def build_ui(self):
19        # Header
20        tk.Label(self.root, text="SALONE KIOSK LEDGER
", font=("Tahoma", 16, "bold"), bg="#0072CE", fg="
white", pady=10).pack(fill=tk.X)
21
22        # Input Frame
23        frame_input = tk.Frame(self.root, bg="#f4f6f9
", pady=15)
24        frame_input.pack()
25
26        tk.Label(frame_input, text="Customer / Item:"
, font=("Tahoma", 10), bg="#f4f6f9").grid(row=0,
column=0, sticky="w", padx=5, pady=5)
27        self.entry_name = tk.Entry(frame_input, width
=30, font=("Tahoma", 10))
28        self.entry_name.grid(row=0, column=1, padx=5
, pady=5)
29
30        tk.Label(frame_input, text="Amount (SLE):",
font=("Tahoma", 10), bg="#f4f6f9").grid(row=1, column
=0, sticky="w", padx=5, pady=5)
31        self.entry_amt = tk.Entry(frame_input, width=

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31 30, font=("Tahoma", 10))
32         self.entry_amt.grid(row=1, column=1, padx=5,
33                               pady=5)
34         tk.Label(frame_input, text="Transaction Type
35 : ", font=("Tahoma", 10), bg="#f4f6f9").grid(row=2,
36 column=0, sticky="w", padx=5, pady=5)
37         self.type_var = tk.StringVar(value="Sale")
38         tk.Radiobutton(frame_input, text="Sale (
39 Income)", variable=self.type_var, value="Sale", bg="#
40 f4f6f9").grid(row=2, column=1, sticky="w")
41         tk.Radiobutton(frame_input, text="Expense (
42 Cost)", variable=self.type_var, value="Expense", bg=
43 "#f4f6f9").grid(row=2, column=1, sticky="e")
44
45         # Buttons Frame
46         frame_btns = tk.Frame(self.root, bg="#f4f6f9"
47 )
48         frame_btns.pack(pady=10)
49
50         tk.Button(frame_btns, text="Save Data",
51 command=self.save_transaction, bg="#1D70B8", fg="
52 white", font=("Tahoma", 9, "bold"), width=12).grid(
53 row=0, column=0, padx=5)
54         tk.Button(frame_btns, text="View Customers",
55 command=self.view_data, bg="#8E44AD", fg="white",
56 font=("Tahoma", 9, "bold"), width=15).grid(row=0,
57 column=1, padx=5)
58         tk.Button(frame_btns, text="Run Report",
59 command=self.generate_report, bg="#1E8449", fg="white
60 ", font=("Tahoma", 9, "bold"), width=12).grid(row=0,
61 column=2, padx=5)
62         tk.Button(frame_btns, text="Exit", command=
63 self.root.quit, bg="#C0392B", fg="white", font=("
64 Tahoma", 9, "bold"), width=8).grid(row=0, column=3,
65 padx=5)
66
67         # Output Display
68         tk.Label(self.root, text="System Output:",
69 font=("Tahoma", 10, "bold"), bg="#f4f6f9").pack(

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50 anchor="w", padx=25, pady=5)
51     self.display_screen = tk.Text(self.root,
    height=15, width=65, font=("Courier", 10), bg="#
    1e1e1e", fg="#00ff00")
52     self.display_screen.pack(padx=20, pady=5)
53     self.print_to_screen("System Ready. Database
    connected.\n")
54
55     def save_transaction(self):
56         name = self.entry_name.get()
57         amount_raw = self.entry_amt.get()
58         trans_type = self.type_var.get()
59
60         # Validate using logic.py
61         is_valid, result = logic.validate_input(name
    , amount_raw)
62
63         if not is_valid:
64             messagebox.showerror("Input Error",
    result)
65             return
66
67         # Save using database.py
68         database.add_transaction(name, result,
    trans_type)
69
70         self.print_to_screen(f"Saved: {name} -> SLE {
    result:.2f} ({trans_type})\n", clear=False)
71         self.entry_name.delete(0, tk.END)
72         self.entry_amt.delete(0, tk.END)
73
74     def view_data(self):
75         """Fetches and displays all data from the CSV
    file."""
76         records = database.get_all_transactions()
77
78         if not records:
79             self.print_to_screen("The database is
    currently empty.\n", clear=True)
80             return
81

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82         output =
            "=====\\n
            "
83         output += f"{'CUSTOMER / ITEM'.ljust(20)} |
{'TYPE'.ljust(10)} | {'AMOUNT'}\\n"
84         output +=
            "=====\\n
            "
85
86         for r in records:
87             output += f"{r['Customer/Item'].ljust(20
)} | {r['Type'].ljust(10)} | SLE {r['Amount']:.2f}\\n
            "
88
89         output +=
            "=====\\n
            "
90         self.print_to_screen(output, clear=True)
91
92         def generate_report(self):
93             records = database.get_all_transactions()
94             if not records:
95                 messagebox.showinfo("Empty", "No data to
calculate.")
96                 return
97
98             sales, expenses, gross, tax, net = logic.
calculate_totals(records)
99             health = logic.evaluate_health(net)
100
101             report = "--- FINANCIAL AUDIT REPORT ---\\n"
102             report += f"Total Sales : SLE {sales:.2f}\\
n"
103             report += f"Total Expenses: SLE {expenses:.
2f}\\n"
104             report += "-----\\n"
105             report += f"Gross Profit : SLE {gross:.2f}\\
n"
106             report += f"Est. Tax (5%) : SLE {tax:.2f}\\n"
107             report += "=====\\n"
108             report += f"NET PROFIT : SLE {net:.2f}\\n"

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109         report += f"STATUS          : {health}\n"
110
111         self.print_to_screen(report, clear=True)
112
113     def print_to_screen(self, text, clear=False):
114         self.display_screen.config(state=tk.NORMAL)
115         if clear:
116             self.display_screen.delete(1.0, tk.END)
117         self.display_screen.insert(tk.END, text)
118         self.display_screen.config(state=tk.DISABLED)
119     )
120 if __name__ == "__main__":
121     app_window = tk.Tk()
122     app = KioskGUI(app_window)
123     app_window.mainloop()
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